

$$h(x) = x^{2/3}(x-4)$$

a. Domain = $\mathbb{R}$

b. Intercepts (0,0), (4,0)

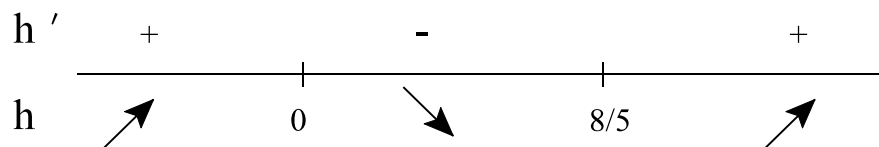
c. Asymptotes

1. Horizontal: None since $\lim_{x \rightarrow \infty} x^{2/3}(x-4) = \infty$, $\lim_{x \rightarrow -\infty} x^{2/3}(x-4) = -\infty$

2. Vertical: None

d. $h'(x) = \frac{5x-8}{3x^{1/3}} \Rightarrow$ C.N. $x = 0$, $x = 8/5$

e.



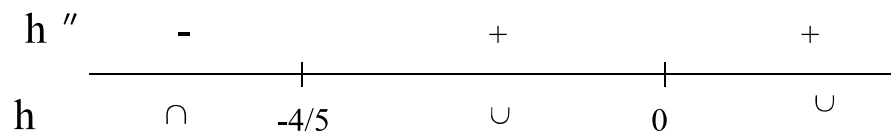
Increasing on $(-\infty, 0) \cup (8/5, \infty)$

Decreasing on $(0, 8/5)$

f. $P(0,0)$ is a relative maximum point on the graph
 $P(8/5, -3.283)$ is a relative minimum point of the graph

g. $h''(x) = \frac{2(5x+4)}{9x^{4/3}} \Rightarrow$ 2nd C.N. $x = -4/5$, $x = 0$

h.



Concave up on $(-4/5, 0) \cup (0, \infty)$

Concave down on $(-\infty, -4/5)$

i. Inflection points: $(-4/5, -4.1365)$

j. Values

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7
y	-26.3	-20.2	-14.6	-9.52	-5	0	-3	-3.17	-2.08	0	2.924	6.604	10.98

k.

